

## trigonometric Substitution

- this gets applied when you have a integral that has one of these

$$\bullet \sqrt{x^2 + a}$$

$$\bullet \sqrt{x^2 - a}$$

$$\bullet \sqrt{a - x^2}$$

- these can either be given the way they are or as a fraction

- "a" can be any number

$$\bullet \sqrt{x^2 + a} \, dx$$

$$x = \sqrt{a} \tan \theta$$

$$dx = \sqrt{a} \sec^2 \theta \, d\theta$$

$$\bullet \sqrt{x^2 - a} \, dx$$

$$x = \sqrt{a} \sec \theta$$

$$dx = \sqrt{a} \sec \theta \tan \theta \, d\theta$$

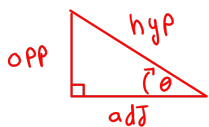
$$\bullet \sqrt{a - x^2} \, dx$$

$$x = \sqrt{a} \sin \theta$$

$$dx = \sqrt{a} \cos \theta \, d\theta$$

- you are given the function and depending on what the function looks like you will substitute the "x" and "dx" accordingly

### review on basic trig



SOH-CAH-TOA

$$\bullet \sin = \frac{\text{opp}}{\text{hyp}}$$

$$\bullet \cos = \frac{\text{adj}}{\text{hyp}}$$

$$\bullet \tan = \frac{\text{opp}}{\text{adj}}$$

$$a^2 + b^2 = c^2 \quad \text{hypotenous}$$

## Practice Problems

$$\int \sqrt{5-x^2} dx$$

$$x = \sqrt{5} \sin \theta \quad dx = \sqrt{5} \cos \theta d\theta$$

$$\int \sqrt{5 - (\sqrt{5} \sin \theta)^2} \cdot \sqrt{5} \cos \theta d\theta$$

$$\int \sqrt{5 - 5 \sin^2 \theta} \cdot \sqrt{5} \cos \theta d\theta$$

$$\int \sqrt{5(1 - \sin^2 \theta)} \cdot \sqrt{5} \cos \theta d\theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\int \sqrt{5 \cdot \cos^2 \theta} \cdot \sqrt{5} \cos \theta d\theta$$

$$\int \sqrt{5} \cdot \cos \theta \cdot \sqrt{5} \cdot \cos \theta$$

$$\int 5 \cos^2 \theta d\theta \quad \cos^2 \theta = \frac{1}{2} - \frac{1}{2} \cos(2\theta)$$

$$5 \int \left[ \frac{1}{2} - \frac{1}{2} \cos(2\theta) \right] d\theta \Rightarrow 5 \int \left[ \frac{1}{2} - \frac{1}{2} \cos(2\theta) \right] d\theta$$

$$u = 2\theta \quad du = 2 d\theta$$

$$5 \int \left[ \frac{1}{2} - \frac{1}{2} \cos(u) \right] \cdot \frac{du}{2}$$

$$\frac{du}{2} = d\theta \quad 5 \left[ \frac{1}{2} \theta - \frac{1}{4} \sin(2\theta) \right]$$

$$\Rightarrow \frac{5}{2} \theta - \frac{5}{4} \sin(2\theta)$$

$$\Rightarrow \frac{5}{2} \theta - \frac{5}{4} \sin \theta \cos \theta$$

- we began in terms of "x" we must end in terms of "x" we will use trig to go from terms of "θ" to terms of "x"

- we already set  $x = \sqrt{5} \sin \theta$  so we can isolate and solve for "sin θ"

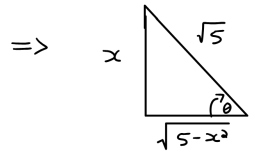
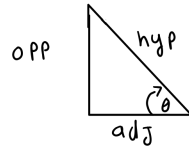
$$\sin \theta = \frac{x}{\sqrt{5}} \quad \frac{\text{opp}}{\text{hyp}}$$

$$a^2 + b^2 = c^2$$

$$a^2 + (x)^2 = (\sqrt{5})^2$$

$$a = \sqrt{5 - x^2}$$

- we can not plug some info from this into the Pythagorean theorem



$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{\sqrt{5-x^2}}{\sqrt{5}}$$

$$\sin \theta = \frac{x}{\sqrt{5}}$$

$$\theta = \sin^{-1}\left(\frac{x}{\sqrt{5}}\right)$$

$$\frac{5}{2} \theta - \frac{5}{4} \sin \theta \cos \theta = \frac{5}{2} \cdot \sin^{-1}\left(\frac{x}{\sqrt{5}}\right) + \frac{5}{4} \cdot \left(\frac{x}{\sqrt{5}}\right) \cdot \frac{\sqrt{5-x^2}}{\sqrt{5}}$$

$$\Rightarrow \frac{5}{2} \sin^{-1}\left(\frac{x}{\sqrt{5}}\right) + \frac{x \sqrt{5-x^2}}{2}$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

- we do this to make it easier to do the next steps

$$\int \frac{1}{x\sqrt{4+x^2}} \cdot dx$$

$$x = 2\tan\theta \quad dx = 2\sec^2\theta d\theta$$

$$\int \frac{1}{(2\tan\theta)\sqrt{4+(2\tan\theta)^2}} \cdot 2\sec^2\theta d\theta$$

$$\int \frac{1}{(2\tan\theta)\sqrt{4+4\tan^2\theta}} \cdot 2\sec^2\theta d\theta$$

$$\int \frac{1}{(2\tan\theta)\sqrt{4(1+\tan^2\theta)}} \cdot 2\sec^2\theta d\theta$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\int \frac{1}{(2\tan\theta)\sqrt{4(\sec^2\theta)}} \cdot 2\sec^2\theta d\theta$$

$$\int \frac{1}{(2\tan\theta) \cdot (\cancel{2}\sec\theta)} \cdot \cancel{2}\sec^2\theta d\theta$$

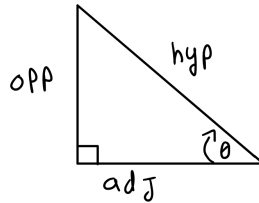
$$\int \frac{\sec\theta}{2\tan\theta} d\theta \Rightarrow \frac{1}{2} \int \frac{1}{\cancel{\cos\theta}} \cdot \frac{\cancel{\cos\theta}}{\sin\theta} d\theta$$

$$\frac{1}{2} \int \frac{1}{\sin\theta} d\theta \Rightarrow \frac{1}{2} \int \csc\theta d\theta$$

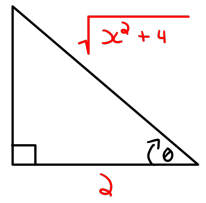
$$\frac{1}{2} \int \csc\theta d\theta \Rightarrow -\frac{1}{2} \ln|\csc\theta + \cot\theta|$$

- We began in terms of "x" we need to end in terms of "x" so we apply trig

$$x = 2\tan\theta \Rightarrow \tan\theta = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$$



$\Rightarrow$



use the pythagreom  
therom to find the  
hypotinous

$$\begin{aligned} a^2 + b^2 &= c^2 \\ (x)^2 + (2)^2 &= c^2 \\ \sqrt{x^2 + 4} &= c \end{aligned}$$

$$\text{SOH-CAH-TOA} \quad \csc\theta = \frac{1}{\sin\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{x}{2} \quad \cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{2}{\sqrt{x^2+4}}$$

$$-\frac{1}{2} \ln|\csc\theta + \cot\theta|$$

$$-\frac{1}{2} \ln\left|\frac{2}{x} + \frac{4}{x\sqrt{x^2+4}}\right|$$